COMPUTER NETWORKS AND HISTORY OF INTERNET

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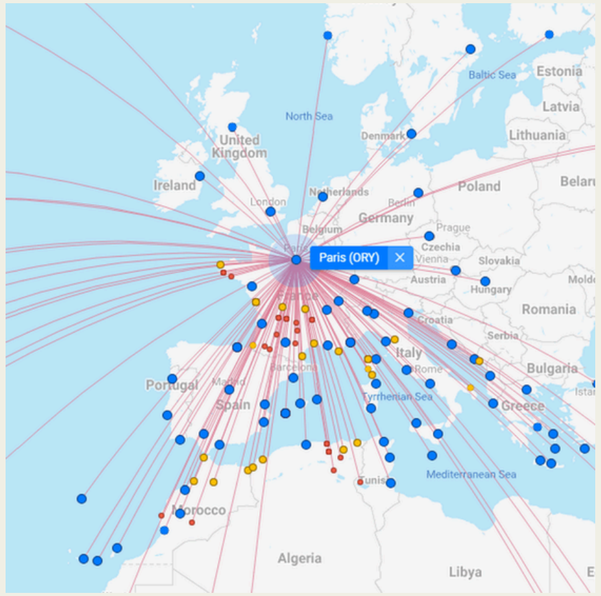
CAMBRIDGE SCHOOL NOIDA

COMPUTER NETWORK AND TYPES

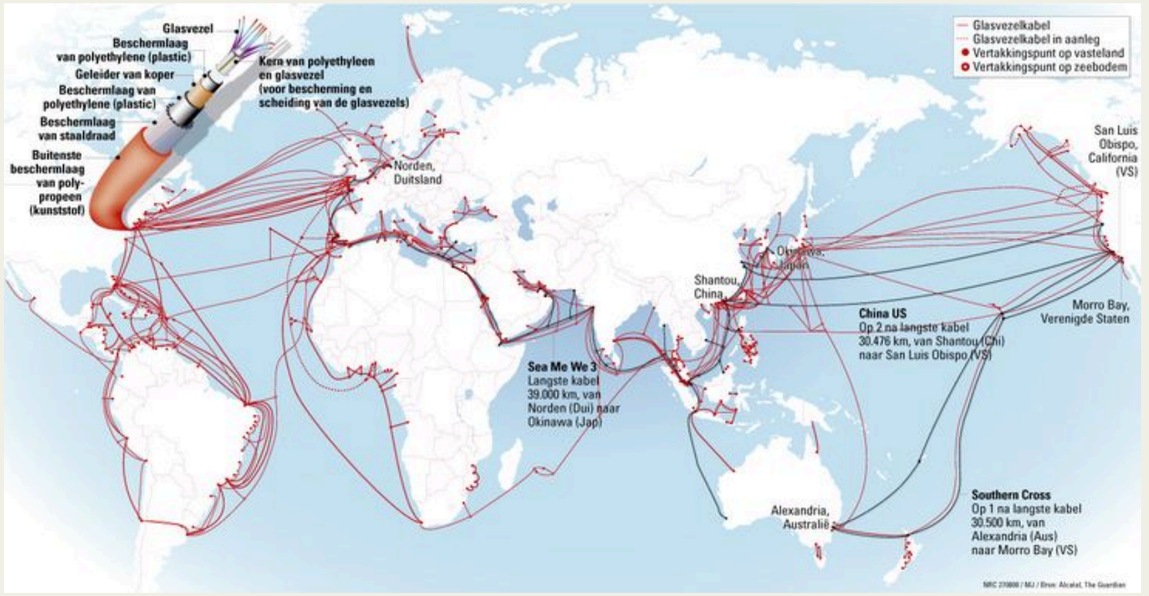
WHAT IS NETWORK ?

A network is a group of two or more similar things or people interconnected with each other.

Example: social networks, mobile networks, networks of computers, and airline, railway, bank and hospital networks.



airline network

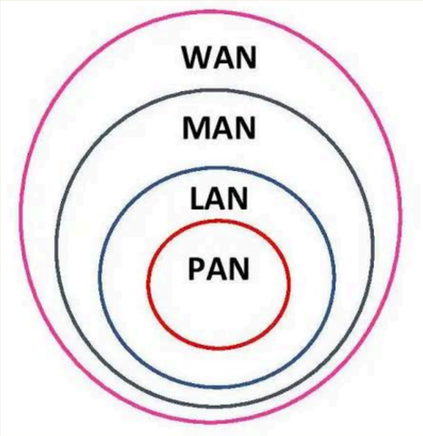


fibre-optics network around the globe

COMPUTER NETWORK

A computer network is an interconnection among two or more computers or computing devices.
It allows them to share data and resources with each other.

- **Personal Area Network (PAN)**
- **Local Area Network (LAN)**
- **Campus Area Network (CAN)** ~~X~~
  - Not studied separately as it is not technically a distinct network type.
- **Metropolitan Area Network (MAN)**
- **Wide Area Network (WAN)**

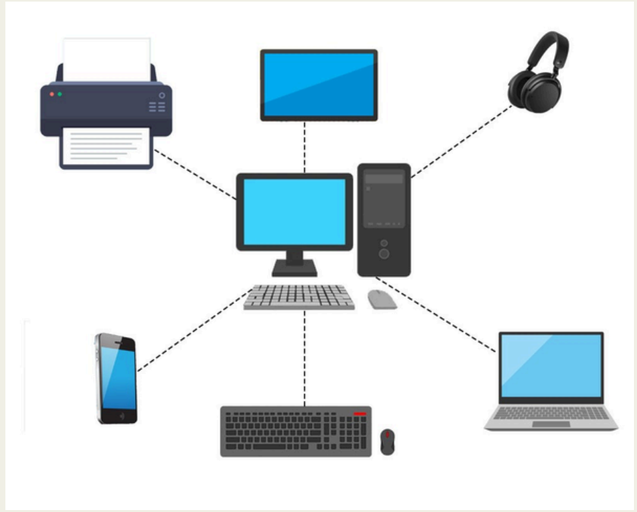


venn-representation

PERSONAL AREA NETWORK (PAN)

It is a network formed by connecting a few personal devices like computers, laptops, mobile phones, smart phones, printers, etc.
Range: approximately 10 metres.
Type: can be wired or wireless.

Example: A PC connected to a keyboard, mouse and monitor wirelessly while connected to headphones and mobile phone wirelessly.



schematic representation of PAN

LOCAL AREA NETWORK(LAN)

It is a network that connects computers, mobile phones, tablets, printers and other devices placed at a limited distance.

Coverage: a single room, a floor, an office, a laboratory, a school, a college or a university campus.

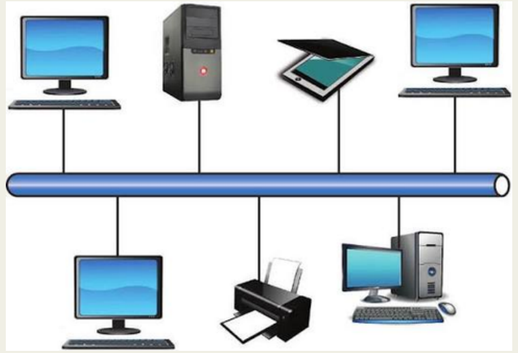
Range: can extend up to 1 km.

Medium: wires, Ethernet cables, fibre optics or Wi-Fi.

Speed: usually 10 Mbps (Ethernet) to 1000 Mbps (Gigabit Ethernet).

Features:

1. Comparatively secure, as only authentic users can access shared resources.
2. Allows printing through a shared printer and uploading or downloading files from a local server.



schematic representation of PAN

METROPOLITAN AREA NETWORK (MAN)

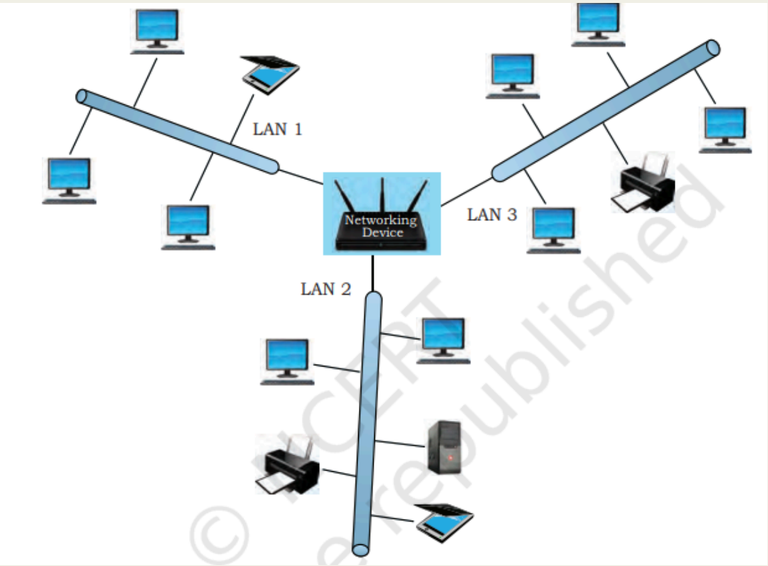
It is an extended form of LAN that covers a larger geographical area such as a city or a town.

Range: can extend up to 30 to 40 km.

Speed: also in Mbps, but considerably lower than LAN.

Formation: sometimes many LANs are connected together to form a MAN.

Example: Cable TV network and cable based broadband internet services.



schematic representation of MAN

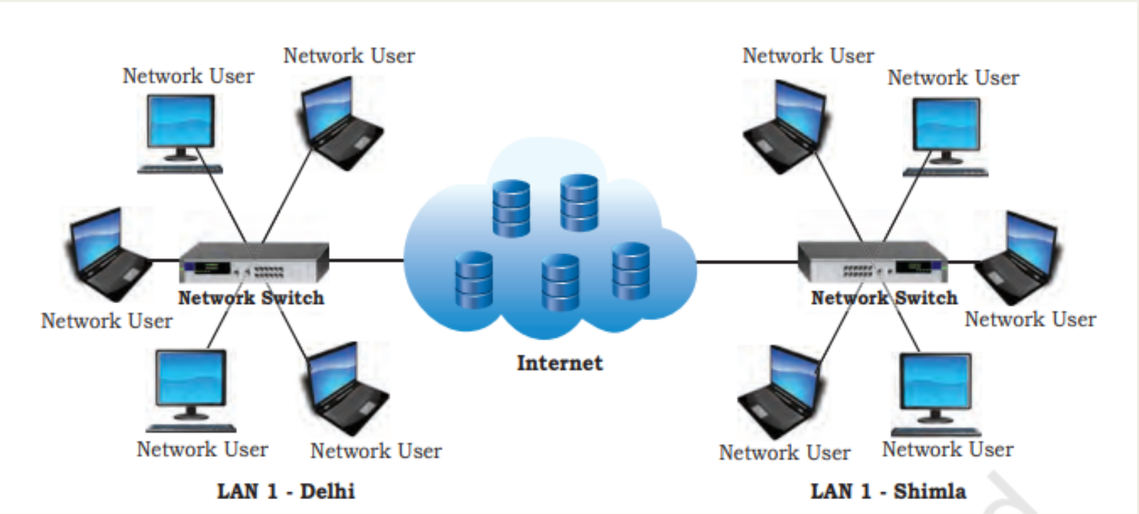
WIDE AREA NETWORK (WAN)

It is a network that connects computers and other LANs and MANs spread across different geographical locations of a country or across countries and continents.

Medium: wired or wireless.

Use: large business, educational and government organisations connect their branches in different locations through a WAN.

Example: The Internet is the largest WAN, connecting billions of computers and smartphones and millions of LANs from different continents.



schematic representation of WAN

HISTORY OF INTERNET

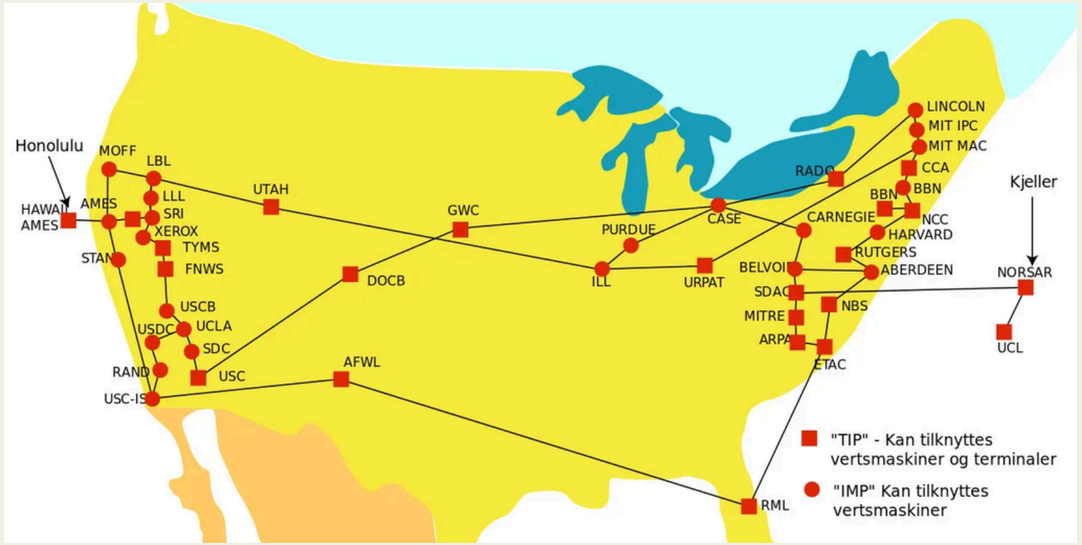
EVOLUTION OF NETWORKING

- In the 1960s, the U.S. Department of Defence commissioned ARPANET (Advanced Research Projects Agency Network).

Aim: to connect academic and research institutions at different places for scientific collaboration.

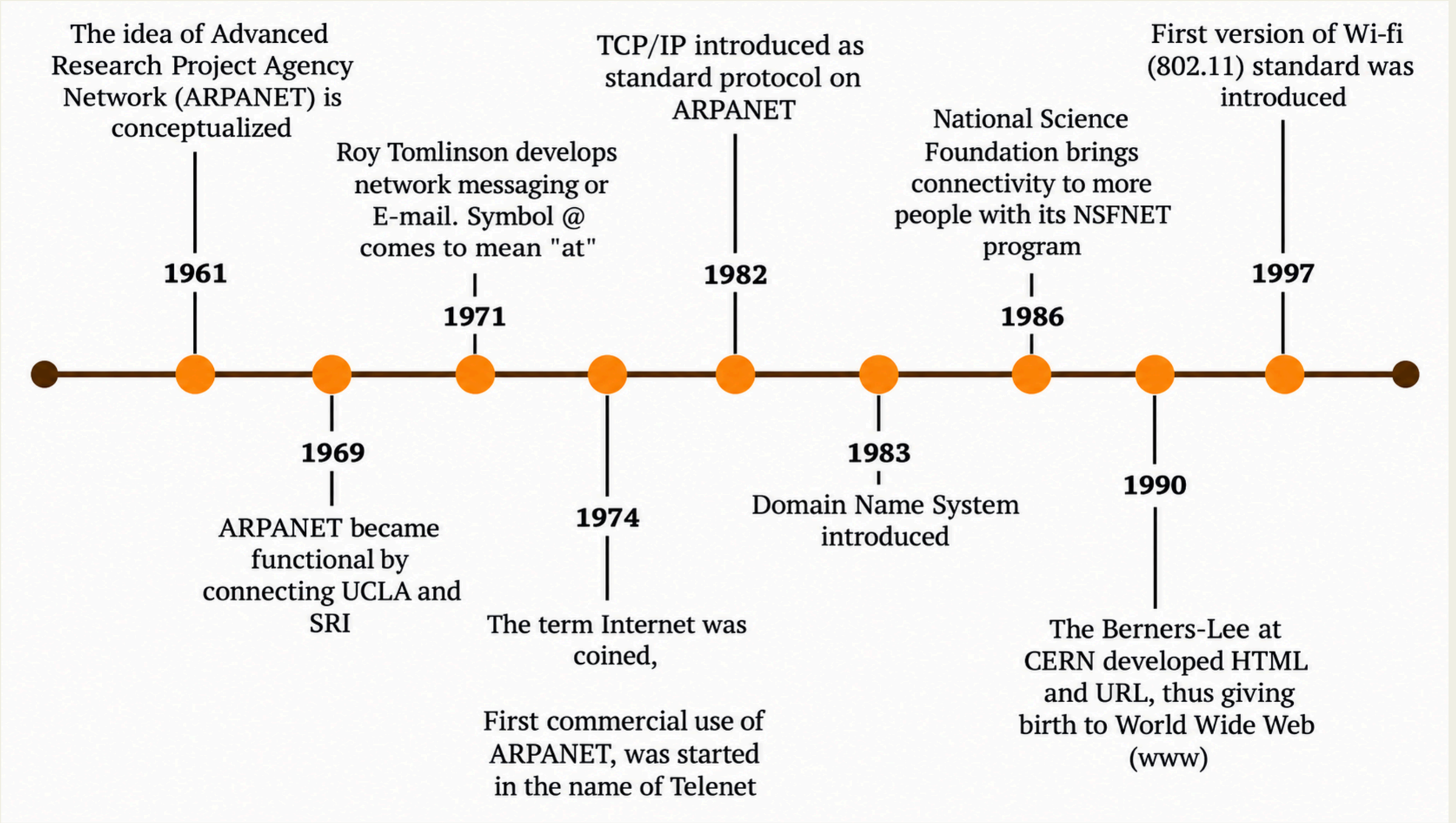
- The first message was sent between the University of California, Los Angeles (UCLA) and Stanford Research Institute (SRI).

- More organisations joined gradually, and many smaller independent networks were formed.

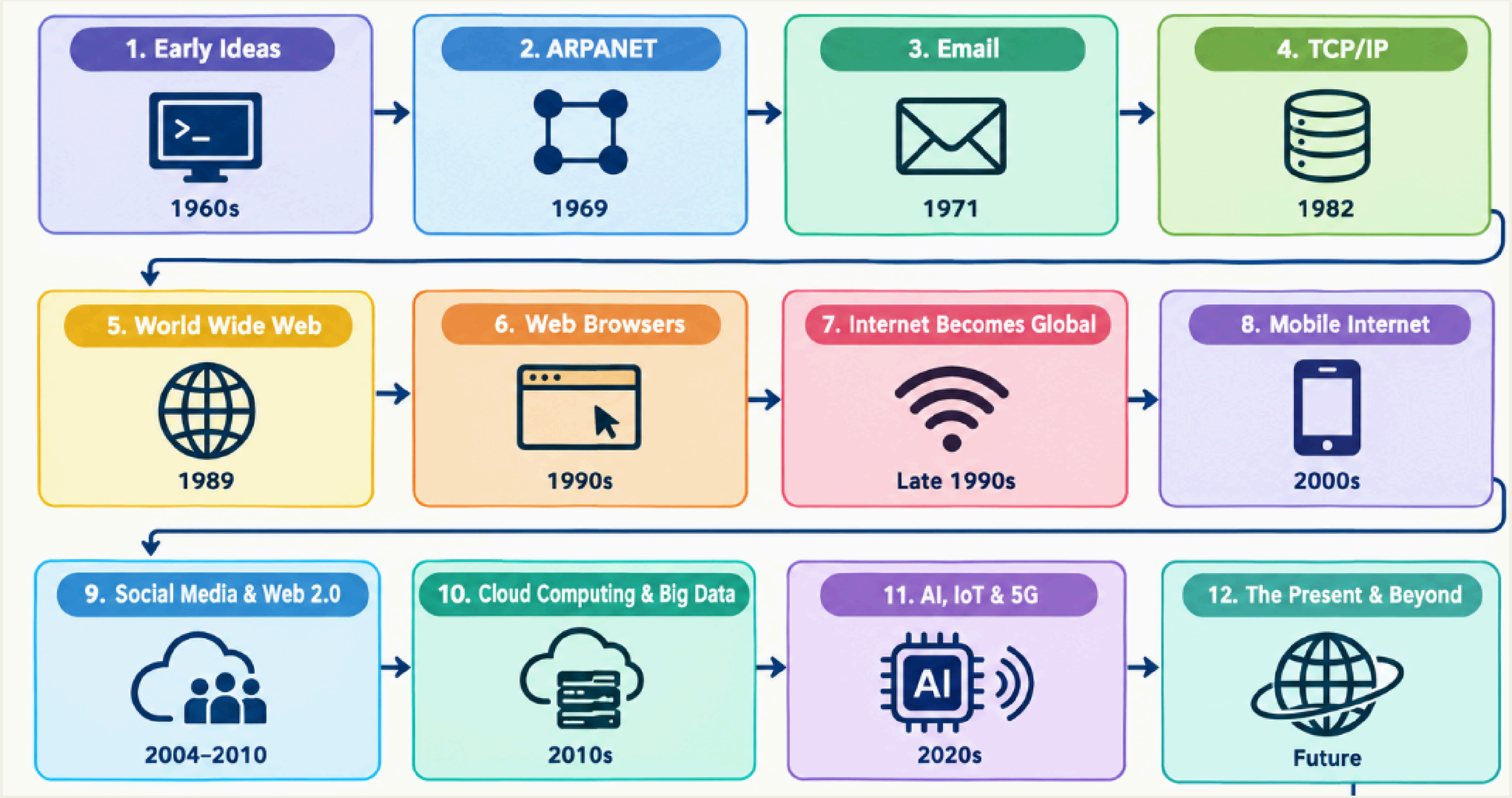


ARPANET during 1960s

TIMELINE


The idea of Advanced Research Project Agency Network (ARPANET) is conceptualized
1961
1969
ARPANET became functional by connecting UCLA and SRI
Roy Tomlinson develops network messaging or E-mail. Symbol @ comes to mean "at"
1971
1974
The term Internet was coined,
First commercial use of ARPANET, was started in the name of Telenet
TCP/IP introduced as standard protocol on ARPANET
1982
1983
Domain Name System introduced
National Science Foundation brings connectivity to more people with its NSFNET program
1986
1990
The Berners-Lee at CERN developed HTML and URL, thus giving birth to World Wide Web (www)
First version of Wi-fi (802.11) standard was introduced
1997

THE DEVELOPMENT



THANK YOU